Class Test
CO4-AT 1

192424385
G. Ram kishore

CSA1402 - Compiler Design

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① Construct three address code for the following
(a<b) and (c<d) or (a>d) then
t = x+y*t
else
z = z+1

Step-1 :- a<b
c<b
a>b

the structure is (a<b) & (c<d) or (a>b)

Step 2 :- L1, L2, L3, L4, L5
Create table

Step 3 :- Generate TAC for condition

L1 : if a<b goto L2
goto L5
L2 : if c<d goto L4
goto L3
L3 : if a>b goto L4
goto L5

Step 4 :- Generate TAC for then part
t = x+y * t

L4 :- t_1 = y*t
t_2 = x+t_1

ii) z = t+1
L5 :-
t_1 = t+1

t_2 = t_1

final :-
L1 = if a<b goto L2
goto L5

L2 : if c<b goto L4
goto L3

L_3: if $a>b$ goto L_4
goto L_5

L_4: $t_1 = y * t$
$t_2 = x + t_1$
$z = t_2$

(Q) Translate $(a<b)$ and $(c<d)$ or $(e<f)$ into TAC statement using back patching

step 1 :- Condition is :- $a<b$
$c<d$
$e<f$

The structure is $(a<b$ and $(c<d)$ or $(e<f)$

Step 2 :- if $a<b$ goto -
goto -
if $c<d$ goto -
goto -
if $e<f$ goto -
goto

Step 3 :- Handle AND
for $(a<b)$ AND $(c<d)$ if $a<b$ is true
$\therefore$ if $a<b$ goto 3

Step 4 :- final backpatched code
if $a<b$ goto 3
goto 5
if $c<d$ goto 7
goto 5
if $e<f$ goto 7
goto 8

True
false

(3) Consider three address code using ... ...

for boolean expression (a>b) or (c< d) or (e<f)

$B \rightarrow B_1 \,||\, B_2$

$B \rightarrow B_1 \,\&\&\, B_2$

$B \rightarrow \,!B_1$

$B \rightarrow E_1 \text{ relop } E_2$

$B \rightarrow$ True

$B \rightarrow$ false

Step 1 :- a>b
c <d
e <f

Assume expression is :-

(a>b) || (c < d) || (e<f)

Step 2 :- Generate three address code for a>b

if a>b goto true,
goto L_1

Step 3 :- c< d
if c<d goto L_1 True
goto L_3

Step 4 :- e<f
if e<f goto L-true
goto L-false

final -three -Address code

if a>b goto Ltrue
goto L_2
if c><d got L-true
goto L_3
if e < f goto Ltrue
goto Lfalse

L-true : if Boolean expression is true
goto LEND

L false :-
if expression is false.

(c) Generate the three ... for following
while (A<C and B>D) do
if A = 1 then C = C+1

L_1 :-
if A<C goto L_2
goto L100

L_2 :- if B>D goto L_3
goto Lend

L_3 :- if A==1 goto L_4
goto L_4

L_{4c} :-
$t_1 = c+1$
$c = t_1$
goto L_1

L_5 :- if A<=0 goto L_6
goto L_1

L_6 :-

$t_2 = A+B$
$A = t_2$
goto